

Deep Research 2026 10 05 Software Assetflow Automation

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Regional ITAD Deep Research 2026 — Software, AssetFlow & Automation

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Purpose: compare mature ITAD software with the needs of a very small Dubbo pilot and define what AssetFlow should and should not automate.

Executive conclusion

Purpose-built ITAD ERP is real and expensive enough that the practitioner's "hundreds per month" comment is credible.

Current public examples:

- RecyclclyERP: £3,000 onboarding plus £54 per user/month, minimum three users. It includes collection/drop-off, CRM, contracts, erase integrations, inventory/demanufacturing, customer portal, e-commerce, finance and fleet features.
- RazorERP: specialised ITAD features include service contracts, fair-market-value pricing, grading devaluation and automated settlements. Capterra Australia lists a starting price of US\$450 per user/month.
- AssetTiger: generic asset management starts at US\$20/month for 500 assets and unlimited users, illustrating how much cheaper generic tracking can be when ITAD-specific workflow is absent.

Sources:

- <https://recyclclyerp.com/pricing>
- <https://www.razorerp.com/itad-software/>
- <https://www.capterra.com.au/software/148503/razorerp>
- <https://www.assettiger.com/pricing>

For a tiny pilot, AssetFlow is rational only if it stays simpler than the business.

1. What mature ITAD ERP actually solves

Commercial ITAD systems commonly combine:

- CRM/customer;
- contracts;
- inbound collections;

- jobs/lots;
- serialized inventory;
- grading;
- FMV;
- wipe integrations;
- component harvesting;
- warehouse locations;
- resale;
- settlement;
- customer portals;
- regulatory documents;
- finance integrations;
- reporting.

That confirms AssetFlow's direction, but it does not mean every feature belongs in Phase 0.

2. Phase 0 minimum system

AssetFlow needs to answer seven questions instantly:

1. What is it?
2. Who did it come from?
3. Where is it?
4. What is its data state?
5. What has happened to it?
6. What happens next?
7. Where did it finally go?

Everything else is secondary.

Minimum fields:

- asset ID;
- serial/IMEI;
- manufacturer/model/category;
- source/customer/job/lot;
- current location;
- status/route;
- data state;
- evidence;
- final destination.

3. Feature priority matrix

P0 — must be reliable now

- authentication/RBAC;
- jobs/lots/assets;
- QR/barcode;
- search/scan;
- location;
- source/customer;
- chain-of-custody events;
- media links;
- data-state gate;
- photos/evidence;
- basic sanitisation record;
- diagnostics;
- grade;
- route;
- downstream;
- basic resale record;
- certificates/evidence export;
- immutable event history.

P1 — build when pilot produces repeated pain

- hands-on time tracking;
- WIP caps;
- ageing/backlog dashboard;
- source-quality metrics;
- repair parts/labour caps;
- charger inventory;
- parts harvesting inventory;
- downstream price book;
- batch scan/move;
- batch label print;
- recall check;
- release authorisation;
- ownership-lock state;
- shipping eligibility.

P2 — wait for actual scale

- automated FMV feeds;

- sophisticated settlement engine;
- customer portal;
- automatic ecommerce listing;
- accounting integration;
- multi-site optimisation;
- SLA workflow;
- external wipe API automation;
- advanced BI.

P3 — avoid until warehouse scale

- RFID;
- robotic/automatic stations;
- complex route engines;
- broad marketplace integrations;
- ML valuation without enough local data.

4. Barcode versus QR versus RFID

QR now

Best Phase 0 choice:

- cheap labels;
- phone scan;
- enough data to encode URL/token;
- human-readable ID printed alongside.

1D barcode

Useful with inexpensive laser scanners and simple asset IDs, but less convenient for embedded URLs and phone scanning.

Data Matrix

Compact and robust, useful in manufacturing, but less familiar to operators and customers.

RFID

Powerful for:

- bulk count;
- no line-of-sight;
- warehouse cycle counting.

Too much infrastructure for a shed pilot.

Trigger:

- repeated evidence that manual scanning is a material labour bottleneck.

5. Recommended label

Human readable:

DEW-2026-000123
Lenovo ThinkPad T14 Gen 2

QR:

`https://assetflow.../assets/<stable-id-or-token>`

Optional:

- data-state warning;
- source/job shorthand.

Do not encode private donor/customer data into the QR.

6. Workflow-rule philosophy

AssetFlow workflow rules should be recommendations with explainable reasons, not opaque automation that silently moves assets.

Example:

```
{
  "conditions": {
    "data_bearing": true,
    "data_state": "UNWIPE_RESTRICTED"
  },
  "action": {
    "route": "SANITISATION",
    "reason": "All data-bearing media must pass the data gate before release."
  }
}
```

Good automation:

- catches missing steps;
- reduces duplicate entry;
- preselects likely workflow;
- warns about caps;
- creates evidence consistency.

Bad automation:

- hides why route changed;
- overrides safety exceptions;
- automatically claims sanitisation success;
- releases assets without operator approval.

7. Workstation model

A workstation is a logical processing profile:

- RECEIVING;
- WIPE;
- DIAGNOSTICS;
- REPAIR;
- GRADING;
- PARTS.

Store:

- workstation ID;
- location;
- tool/profile;
- operator;
- active state.

A physical bench can serve multiple profiles at Phase 0.

8. Data sanitisation integration maturity

Stage 1

Manual run + upload raw report.

Stage 2

Parse report into:

- asset/media ID;
- tool;
- method;
- start/end;
- result;
- validation.

Stage 3

Controlled launch hook to external tool.

Stage 4

API/integration with signed results.

Stage 5

Batch orchestration and exception queue.

Do not let AssetFlow itself claim to wipe drives merely because it stores a wipe record.

9. Fair-market value and settlements

RazorERP's public ITAD positioning shows why mature platforms connect:

- base FMV;
- grade deductions;
- contracts;
- settlement.

For Phase 0, keep this simpler.

Suggested record:

- model/spec;
- reference price source/date;
- expected grade;
- expected sale range;
- actual grade;
- actual sale;
- fees;
- parts;
- labour minutes;
- net recovery.

After enough records, AssetFlow can derive local FMV.

Do not use AI-generated price estimates as settlement truth without market evidence.

10. Charger data model

Suggested fields:

```
charger_id
manufacturer
model
connector_type
voltage
amps
watts
polarity
usb_c_pd
oem_status
condition
tested_status
compatible_families
location_id
source_asset_id
sale_status
notes
```

High practical value because accessory clutter was a real practitioner pain point.

11. Parts inventory model

Fields:

```
part_id
parent_asset_id
part_type
manufacturer
part_number
spec
condition
test_status
data_bearing
location
compatible_models
harvested_at
used_in_asset_id
sold_at
downstreamed_at
```

Rule:

No unidentified data-bearing media in parts stock.

12. Backlog dashboard

Add four headline measures:

```
inbound_this_week
closed_this_week
current_wip
assets_over_age_limit
```

Warn when:

- inbound > closed for two weeks;
- any zone over cap;
- unwiped asset ageing exceeds threshold;
- repair WIP has no next action/review date.

13. Data quality rules

High value:

- serial uniqueness where applicable;
- required source/job before intake complete;
- data-bearing defaults to restricted;
- release requires final data state;
- current location cannot be null after intake;
- route change creates event;
- media removal creates parent-child link;
- deletion of audit history prohibited.

These controls create more value than cosmetic automation.

14. Build-versus-buy trigger

Continue AssetFlow while:

- one/few operators;
- workflow still changing;
- custom regional process matters;
- hosting/development cost is low;
- it reliably protects custody/data state.

Reassess commercial ERP when:

- multiple staff/sites;
- customer portals become mandatory;
- finance/settlement complexity grows;
- major ITAD contracts demand mature reporting;
- integration maintenance consumes more labour than licence cost;
- uptime/support expectations exceed hobby/custom software tolerance.

15. Total-cost test

For any automation feature:

annual benefit
= minutes saved per transaction
× transactions/year
× loaded labour rate
+ expected error/risk reduction

annual cost
= build time
+ hosting/licence
+ maintenance
+ training
+ failure/rework

If benefit cannot be articulated, defer it.

16. Recommended AssetFlow backlog from this research

Immediate

1. backlog/WIP dashboard;
2. release-authorisation gate;
3. ownership-lock field;
4. hands-on minutes;
5. repair cap fields;
6. recall check;

7. shipping/battery eligibility;
8. charger inventory;
9. downstream price book;
10. evidence retention class.

Next

11. parts harvesting inventory;
12. batch scan/move;
13. source-quality score;
14. route ageing;
15. settlement reconciliation;
16. sanitisation report parser;
17. customer-facing manifest export.

Later

18. external data-erasure API;
19. FMV automation;
20. accounting integration;
21. customer portal;
22. RFID.

17. Security principle

AssetFlow is an operational system containing:

- serials/IMEIs;
- customer/source contacts;
- chain-of-custody;
- wipe reports;
- prices/margins;
- buyer details.

Therefore:

- keep production records behind authentication;
- RLS/least privilege;
- no service-role secret in browser;
- minimise personal data;
- protect evidence;
- back up database;
- keep audit history;
- do not expose operational data in the public GitHub repository.

The public repository can contain schema/process documentation, not live private records.

18. Conclusion

AssetFlow should not try to beat a mature ERP feature-for-feature.

Its competitive advantage in Phase 0 is:

- exact fit;
- low cost;
- fast iteration;
- clear custody;
- regional workflow;
- evidence-first design.

If it becomes slower than the physical work, simplify it.